

Transforming Quadratic Functions from Vertex Form to General Form

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A quadratic function may be written in different but equivalent forms.

Vertex Form

The **vertex form** is

$$y = a(x - h)^2 + k$$

Basic Quadratic Form

The basic quadratic form is

$$y = ax^2$$

Vertical Translation Form

The **vertical translation form** is

$$y = ax^2 + b$$

Horizontal Translation Form

The **horizontal translation form** is

$$y = a(x - b)^2$$

Standard Form

The **standard form** is

$$y = ax^2 + bx + c$$

Vertex Form to Standard Form

To transform from **vertex form** to **standard form**:

1. Expand $(x - h)^2 = x^2 - 2hx + h^2$.
2. Multiply by a .
3. Combine like terms.

Examples

Example 1

Rewrite $y = 2(x - 3)^2 + 5$ in standard form.

$$y = 2(x^2 - 6x + 9) + 5$$

$$y = 2x^2 - 12x + 18 + 5$$

$$y = 2x^2 - 12x + 23$$

Example 2

Rewrite $y = -(x + 4)^2 + 1$ in standard form.

$$y = -(x^2 + 8x + 16) + 1$$

$$y = -x^2 - 8x - 16 + 1$$

$$y = -x^2 - 8x - 15$$

Example 3

Rewrite $y = \frac{1}{2}(x - 2)^2 + 3$ in standard form.

$$y = \frac{1}{2}(x^2 - 4x + 4) + 3$$

$$y = \frac{1}{2}x^2 - 2x + 2 + 3$$

$$y = \frac{1}{2}x^2 - 2x + 5$$

Practice Exercises

Practice 1

Rewrite in the indicated form.

$$y = 3(x - 0)^2 + 0; \quad y = ax^2$$

Practice 2

Rewrite in the indicated form.

$$y = 2(x - 0)^2 + 7; \quad y = ax^2 + b$$

Practice 3

Rewrite in the indicated form.

$$y = -4(x + 5)^2; \quad y = a(x - b)^2$$

Practice 4

Rewrite in the indicated form.

$$y = (x - 4)^2 + 6; \quad y = ax^2 + bx + c$$

Practice 5

Rewrite in the indicated form.

$$y = 2(x + 3)^2 - 5; \quad y = ax^2 + bx + c$$

Practice 6

Rewrite in the indicated form.

$$y = -3(x - 2)^2 + 1; \quad y = ax^2 + bx + c$$

Practice 7

Rewrite in the indicated form.

$$y = 5(x - 1)^2; \quad y = ax^2 + bx + c$$

Practice 8

Rewrite in the indicated form.

$$y = -2(x + 4)^2 - 3; \quad y = ax^2 + bx + c$$

Practice 9

Rewrite in the indicated form.

$$y = \frac{1}{2}(x - 6)^2 + 4; \quad y = ax^2 + bx + c$$

Practice 10

Rewrite in the indicated form.

$$y = -\frac{3}{2}(x + 2)^2 + 5; \quad y = ax^2 + bx + c$$

Activities

Activity 1

Rewrite in the indicated form.

$$y = -2(x - 0)^2 + 0; \quad y = ax^2$$

Activity 2

Rewrite in the indicated form.

$$y = 5(x - 0)^2 - 4; \quad y = ax^2 + b$$

Activity 3

Rewrite in the indicated form.

$$y = 3(x + 7)^2; \quad y = a(x - b)^2$$

Activity 4

Rewrite in the indicated form.

$$y = (x + 5)^2 - 8; \quad y = ax^2 + bx + c$$

Activity 5

Rewrite in the indicated form.

$$y = 4(x - 2)^2 + 3; \quad y = ax^2 + bx + c$$

Activity 6

Rewrite in the indicated form.

$$y = -2(x + 1)^2 - 6; \quad y = ax^2 + bx + c$$

Activity 7

Rewrite in the indicated form.

$$y = 6(x + 3)^2; \quad y = ax^2 + bx + c$$

Activity 8

Rewrite in the indicated form.

$$y = -5(x - 4)^2 + 2; \quad y = ax^2 + bx + c$$

Activity 9

Rewrite in the indicated form.

$$y = \frac{3}{2}(x - 2)^2 - 1; \quad y = ax^2 + bx + c$$

Activity 10

Rewrite in the indicated form.

$$y = -\frac{1}{2}(x - 8)^2 + 7; \quad y = ax^2 + bx + c$$

Thank you!